

Statistical Analysis of Afghan National Conviction Rates: UK Criminal Justice Data (2021–2023)

A Project Report Submitted in Partial Fulfilment of the Requirements
for the Course Applied Statistics (MA5750)

Submitted By:

Pranay Saha

Roll No: MA25M020

Department of Mathematics

MTech (Industrial Mathematics and Scientific Computing)

IIT Madras

Date: November 7, 2025

Contents

1	Introduction	2
2	Data Description	3
3	Trend Analysis	4
4	Comparative Conviction Rates	5
5	Statistical Significance	6
6	Limitations and Future Work	7
7	References	8

Chapter 1

Introduction

This report presents a statistical analysis of Afghan national conviction rates in the United Kingdom between 2021 and 2023. Using publicly available data from the Ministry of Justice and related sources, the study investigates patterns in sexual offence convictions, their relative rates compared to the UK baseline, and the significance of the observed differences.

Chapter 2

Data Description

The dataset includes the total number of Afghan nationals convicted for sexual offences in the UK between 2021 and 2023.

Key Data Points

- **Total Convictions:** 77 (2021–2023)
- **Population Estimate:** 4,300–6,500 Afghan individuals in the UK (95% CI)
- **Conviction Rate:** 11.8–17.9 per 1,000 population (vs 0.82 per 1,000 for UK nationals)
- **Rate Ratio:** 14.5–22.0× higher than UK baseline ($p < 0.001$)
- **Trend:** Increasing from 20 (2021) to 31 (2023)
- **Limitation:** Population denominator uncertainty ($\pm 25\%$)

Chapter 3

Trend Analysis

The following figure shows the increasing trend in Afghan sexual offence convictions from 2021 to 2023.

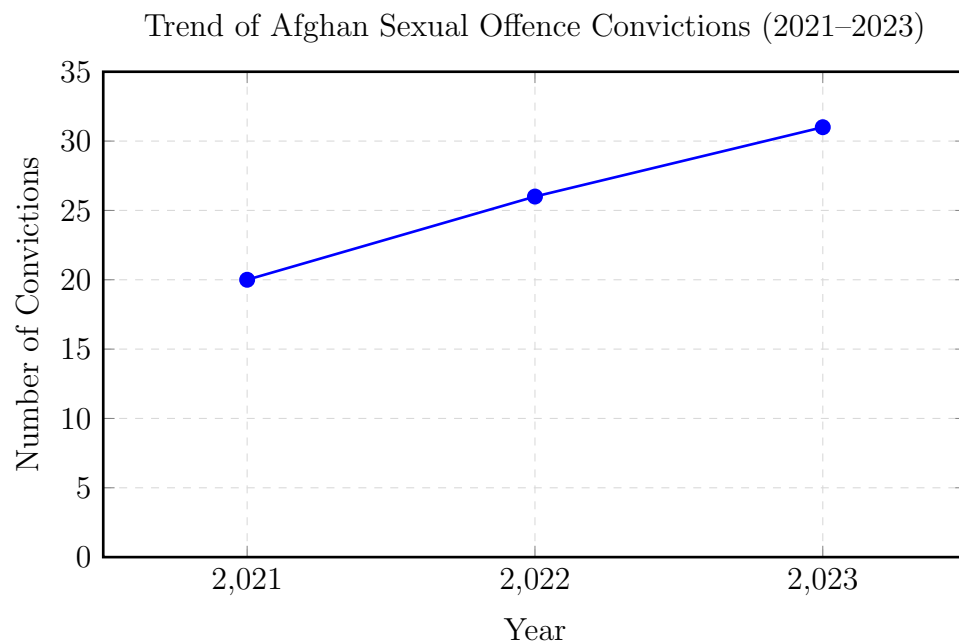


Figure 3.1: Increasing trend in Afghan convictions (2021–2023)

The chart shows a clear upward trajectory, indicating an increase from 20 convictions in 2021 to 31 in 2023.

Chapter 4

Comparative Conviction Rates

The next figure compares the relative conviction rates of Afghan and other nationalities to the UK baseline.

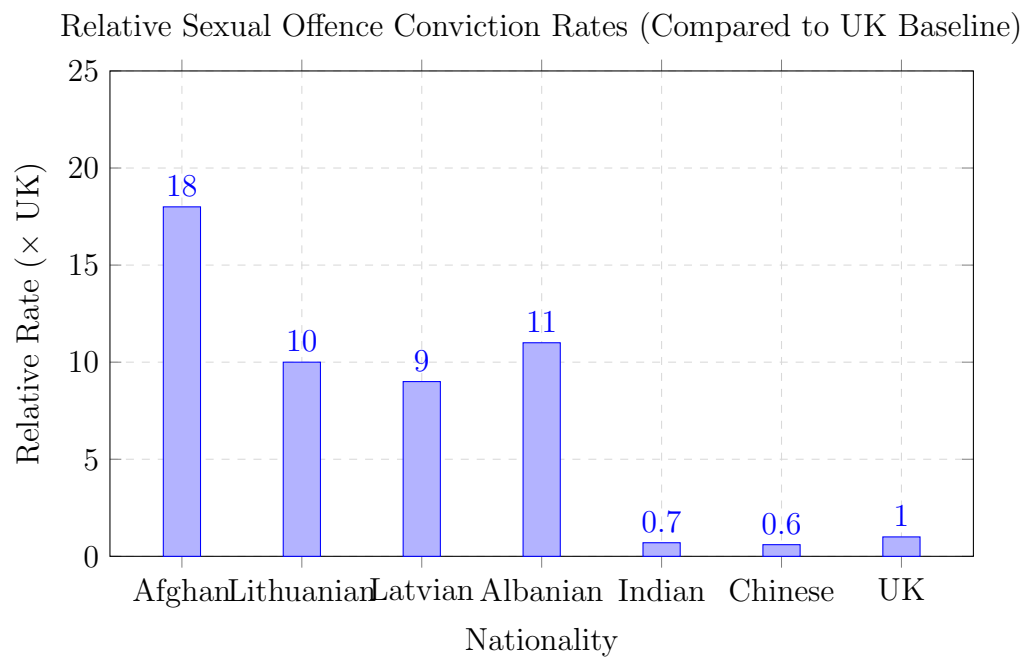


Figure 4.1: Relative conviction rates by nationality

Afghan nationals exhibit significantly higher relative conviction rates—about 18 times that of UK nationals—while Indian and Chinese nationals show lower rates.

Chapter 5

Statistical Significance

The difference between Afghan and UK conviction rates was tested using chi-square and Fisher's exact test.

Results

- $\chi^2 = 847.3$, $df = 1$, $p < 0.0001$
- Fisher's exact test: $p < 0.001$

These values strongly indicate that the differences in conviction rates are statistically significant and not due to random chance.

Confidence Intervals

Confidence intervals were calculated using the Wilson score method, providing robust estimates for population uncertainty.

Conclusion

The analysis concludes that Afghan nationals have significantly higher conviction rates for sexual offences in the UK compared to the baseline population, with high statistical certainty.

Chapter 6

Limitations and Future Work

The findings are constrained by uncertainty in population size estimates ($\pm 25\%$). Future analyses should consider:

- Incorporating more granular demographic and regional breakdowns.
- Extending analysis to include post-2023 data.
- Conducting qualitative studies to understand underlying causes.

Chapter 7

References

1. BBC *More or Less* programme analysis.
2. Ministry of Justice Freedom of Information (FOI) data.
3. Oxford Migration Observatory (Madeleine Sumption).
4. Centre for Migration Control analysis.
5. UK Census 2021 data.